Part A

Question 1:

a)

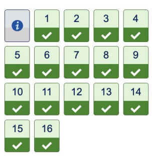
i.

Section 1 and 2 are core.

ii.

Section 3 is non-core.

b)



Practice quiz 2

Summary of attempt

Question	Status	Marks
1	Correct	2.00
2	Correct	1.50
3	Correct	0.89
4	Correct	1.00
5	Correct	1.00
6	Correct	0.75
7	Correct	1.00
8	Correct	1.93
9	Correct	2.00
10	Correct	1.00
11	Correct	2.00
12	Correct	2.00
13	Correct	1.00
14	Correct	1.00
15	Correct	0.50
16	Correct	0.50

Question 2:

$$G = \{a + b\sqrt{-5} : a, b \in \mathbb{Z}\}$$

and $+_G$ be

$$(a + a') + (b + b')\sqrt{-5}$$

and a subgroup of G

$$H = \{a + b\sqrt{-5} : a, b \in 4\mathbb{Z}\}$$

a)
i.
The identity element, e, of G is, $0 + 0\sqrt{-5}$.

ii.
An element of H that is not the identity is, $4 + 8\sqrt{-5}$.

iii.
An element of G that is not in H is, $3 + 5\sqrt{-5}$.

iv.
The sum, under $+_G$, of these two elements are,

$$\begin{aligned}(4 + 8\sqrt{-5}) + (3 + 5\sqrt{-5}) &= (4 + 3) + (8 + 5)\sqrt{-5} \\ &= 7 + 13\sqrt{-5}\end{aligned}$$

b)
G1 Closure:

Let,

$$x = a + b\sqrt{-5} \quad \text{and } y = a' + b'\sqrt{-5}$$

then

$$x + y = (a + a') + (b + b')\sqrt{-5}$$

and as $\mathbb{Z}$ is closed under addition $a + a' \in \mathbb{Z}$ and $b + b' \in \mathbb{Z}$, Hence, $x + y \in G$.

G2 Identity element:

Let,

$$e = 0 + 0\sqrt{-5}$$

and

$$x = a + b\sqrt{-5}$$

then

$$\begin{aligned} x + e &= (a + 0) + (b + 0)\sqrt{-5} \\ &= a + b\sqrt{-5} \\ &= x \end{aligned}$$

G3 Inverse element:

Let,

$$x = a + b\sqrt{-5}$$

then

$$\begin{aligned} x + (-x) &= (a + -a) + (b + -b)\sqrt{-5} \\ &= 0 + 0\sqrt{-5} \\ &= e \end{aligned}$$

G4 Associativity:

The associativity condition holds for addition in $\mathbb{Z}$,

Hence $(G, +_G)$ is a group.

c)
Using Proposition 2.5; Subgroup criteria, HB Chapter 5 p26.

$$x, y \in H \implies x - y \in H$$

Let,

$$x = 4m + 4n\sqrt{-5} \quad \text{and } y = 4p + 4q\sqrt{-5}$$

then

$$\begin{aligned} x - y &= (4m - 4p) + (4n - 4q)\sqrt{-5} \\ &= 4(m - p) + 4(n - q)\sqrt{-5} \end{aligned}$$

As $m - p \in \mathbb{Z}$ and $n - q \in \mathbb{Z}$, then $x - y \in H$. Hence, H is a subgroup of G.

d)
The number of distinct cosets of H in G is 16.

- $a \pmod 4$ has 4 possible values; $0, 1, 2, 3$ in $4\mathbb{Z}$.
- $b \pmod 4$ has 4 possible values; $0, 1, 2, 3$ in $4\mathbb{Z}$.

Thus, by the counting principle, the total number of cosets is $4 \times 4 = 16$.

e)
The order of the element $1 + \sqrt{-5} + H \in \frac{G}{H}$ is 4.

$$\left(1 + \sqrt{-5}\right) + H, \quad 2\left(1 + \sqrt{-5}\right) + H, \quad 3\left(1 + \sqrt{-5}\right) + H, \quad 4\left(1 + \sqrt{-5}\right) + H = H$$

Question 3:

$$M = \begin{pmatrix} a, 0 \\ b, c \end{pmatrix} : a, b, c \in \mathbb{Z}_5$$

a)
The additive inverse is

$$\begin{pmatrix} -a, -0 \\ -b, -c \end{pmatrix}$$

b)
Let

$$\varphi : M \to \mathbb{Z}_5$$

be the map that takes a matrix in M to the lower left entry in M.

Proof. To show that φ is a homomorphism, let

$$X = \begin{pmatrix} a, 0 \\ b, c \end{pmatrix} \quad \text{and} \quad Y = \begin{pmatrix} d, 0 \\ e, f \end{pmatrix}$$

then

$$\begin{aligned} \varphi(X + Y) &= \varphi \begin{pmatrix} a + d, 0 + 0 \\ b + e, c + f \end{pmatrix} \\ &= b + e \end{aligned}$$

and

$$\varphi(X) + \phi(Y) = b + e$$

Thus,

$$\varphi(X + Y) = \varphi(X) + \varphi(Y)$$

Hence, φ is a homomorphism. $\square$

c)

The kernel of φ is,

$$Ker\,(\varphi) = \begin{pmatrix} a, 0 \\ 0, c \end{pmatrix} : a, c \in \mathbb{Z}_5$$

because the lower-left entry must be 0.

d)

A group that is isomorphic to the quotient group $M/Ker(\varphi)$ is $\mathbb{Z}_5$

Question 4:

$$G = \mathbb{Z}_{101} \times \mathbb{Z}_{50}$$

a)

As 101 and 50 are coprime (Corollary 2.12, HB Chapter 6, pg.32),

$$\mathbb{Z}_{101} \times \mathbb{Z}_{50} \cong \mathbb{Z}_{5050}$$

The order of G is $101 \times 50 = 5050$.

i.

The order of the element $(25, 1)$ is;

$$\frac{101}{\text{hcf}(101, 25)} = 101$$
$$\frac{50}{\text{hcf}(50, 1)} = 50$$

Thus, the order of (25,1) is

$$\text{lcm}(101, 50) = 5050$$

Hence $(25, 1)$ is a generator of G.

ii.

The order of the element $(1, 25)$ is;

$$\frac{101}{\text{hcf}(101, 1)} = 101$$
$$\frac{50}{\text{hcf}(50, 25)} = 2$$

Thus, the order of (1,25) is

$$\text{lcm}(101, 2) = 202$$

Hence $(1, 25)$ is not a generator of G.

b)
i.
First group

$$\mathbb{Z}_{600} \times \mathbb{Z}_{49}$$

The order of this group is $600 \times 49 = 29400$.

Using Corollary 2.12 and Theorem 2.13, HB Chapter 6, p32,

$$\begin{aligned}\mathbb{Z}_{600} &\cong \mathbb{Z}_8 \times \mathbb{Z}_3 \times \mathbb{Z}_{25} \\ \mathbb{Z}_{49} &\cong \mathbb{Z}_{49}\end{aligned}$$

Hence,

$$\mathbb{Z}_{600} \times \mathbb{Z}_{49} \cong \mathbb{Z}_8 \times \mathbb{Z}_3 \times \mathbb{Z}_{25} \times \mathbb{Z}_{49}$$

Second group

$$\mathbb{Z}_2 \times \mathbb{Z}_{12} \times \mathbb{Z}_{175} \times \mathbb{Z}_7$$

The order of this group is $2 \times 12 \times 175 \times 7 = 29400$.

Using Corollary 2.12 and Theorem 2.13, HB Chapter 6, p32,

$$\begin{aligned}\mathbb{Z}_2 &\cong \mathbb{Z}_2 \\ \mathbb{Z}_{12} &\cong \mathbb{Z}_4 \times \mathbb{Z}_3 \\ \mathbb{Z}_{175} &\cong \mathbb{Z}_{25} \times \mathbb{Z}_7 \\ \mathbb{Z}_7 &\cong \mathbb{Z}_7\end{aligned}$$

Therefore a decomposition as a direct product of cyclic groups of prime power order is,

$$\mathbb{Z}_2 \times \mathbb{Z}_4 \times \mathbb{Z}_3 \times \mathbb{Z}_{25} \times \mathbb{Z}_7 \times \mathbb{Z}_7$$

Thus (By Theorem 2.13, HB Chapter 6, p32),

$$\mathbb{Z}_2 \times \mathbb{Z}_{12} \times \mathbb{Z}_{175} \times \mathbb{Z}_7 \cong \mathbb{Z}_{14} \times \mathbb{Z}_{2100}$$

Third group

$$\mathbb{Z}_2 \times \mathbb{Z}_2 \times \mathbb{Z}_6 \times \mathbb{Z}_{1225}$$

The order of this group is $2 \times 2 \times 6 \times 1225 = 29400$.

Using Corollary 2.12 and Theorem 2.13, HB Chapter 6, p32,

$$\mathbb{Z}_2 \cong \mathbb{Z}_2$$
$$\mathbb{Z}_6 \cong \mathbb{Z}_2 \times \mathbb{Z}_3$$
$$\mathbb{Z}_{1225} \cong \mathbb{Z}_{25} \times \mathbb{Z}_{49}$$

Thus (By Theorem 2.13, HB Chapter 6, p32),

$$\mathbb{Z}_2 \times \mathbb{Z}_2 \times \mathbb{Z}_6 \times \mathbb{Z}_{1225} \cong \mathbb{Z}_2 \times \mathbb{Z}_2 \times \mathbb{Z}_2 \times \mathbb{Z}_3 \times \mathbb{Z}_{25} \times \mathbb{Z}_{49}$$

ii.

By Theorem 2.11, HB Chapter 6, p32, a direct product is cyclic if and only if the component orders are coprime.

- First group; $\mathbb{Z}_{600} \times \mathbb{Z}_{49}$; $\mathrm{hcf}(600, 49) = 1$ so it is cyclic.
- Second group; $\mathbb{Z}_2 \times \mathbb{Z}_{12} \times \mathbb{Z}_{175} \times \mathbb{Z}_7$; $\mathrm{hcf}(2, 12) = 2$ so it is not cyclic.
- Third group; $\mathbb{Z}_2 \times \mathbb{Z}_2 \times \mathbb{Z}_6 \times \mathbb{Z}_{1225}$; $\mathrm{hcf}(2, 6) = 2$ so it is not cyclic.

Therefore only the group $\mathbb{Z}_{600} \times \mathbb{Z}_{49}$ is cyclic.

c)

Since

$$1144 = 2^3 \times 11 \times 13$$

and, Using Corollary 2.12, HB Chapter 6, p32,

$$\mathbb{Z}_{11} \times \mathbb{Z}_{13} \cong \mathbb{Z}_{143}$$

Two possible (non-isomorphic) non-cyclic abelian groups of order 1144 are,

$$\mathbb{Z}_{1144} \cong \mathbb{Z}_4 \times \mathbb{Z}_2 \times \mathbb{Z}_{143}$$
$$\mathbb{Z}_{1144} \cong \mathbb{Z}_2 \times \mathbb{Z}_2 \times \mathbb{Z}_2 \times \mathbb{Z}_{143}$$

They are non-isomorphic because their 2-power components differ.

Question 5:

$$D_{50} = \langle r, s \mid r^{50} = s^2 = e, sr = r^{49}s \rangle$$

a)
Let $H = \{r^{2i}s^j : i = 0, \ldots, 24, j = 0, 1\}$

Using Proposition 2.4, HB p25.

Proof.
Non-empty:

Let $i = 0, j = 0$

$$r^0 s^0 = e \in H$$

Thus, H is non-empty.

Closure $x, y \in H \implies x^{-1}y \in H$:

Let

$$x = r^{2a}s^b \text{ and } y = r^{2c}s^d, \quad a, c \in \{0, \ldots, 24\}, \quad b, d \in \{0, 1\}$$

then

$$x^{-1} = s^b r^{-2a}$$

Since $sr = r^{-1}s$, we can move s past powers of r.

$$x^{-1}y = r^{-2a}s^b \cdot r^{2c}s^d$$

$$= r^{-2a} \cdot s^{b+d} \cdot r^{2c}$$

Let $j = 0$,

$$x^{-1}y = r^{-2a} \cdot r^{2c}$$

$$= r^{2(c-a)} \in H$$

Let $j = 1$,

$$x^{-1}y = r^{-2a}s \cdot r^{2c}s$$

Again since $sr = r^{-1}s$, we can move s past powers of r.

$$= r^{-2a} \cdot r^{2c} \cdot s^2$$

As $s^2 = e$,

$$= r^{2(c-a)} \in H$$

Hence, H is a subgroup of D_{50}.

$\square$

b)

The order of the group D_{50} is 100. The order of the subgroup H is 50.

Hence the index of H in D_{50} is

$$\frac{\mid D_{50} \mid}{\mid H \mid} = \frac{100}{50} = 2$$

As the index of H is 2, its left and right cosets are the same, hence H is a normal subgroup of D_{50}, by Proposition 3.7, HB Chapter 6, p27.

c)

Let k be the subgroup generated by r^{25}.

The order of k is 2, as

$$(r^{25})^1 = r^{25}$$
$$(r^{25})^2 = r^{50} = e$$

Which is isomorphic to $\mathbb{Z}_2$ which also has order 2.

d)
Given $D_{50}\langle sr = r^{49}s = r^{-1}s\rangle$;

$$Z(H) = \{\, a \in H : ah = ha \,\forall\, h \in H\}$$

$$\begin{aligned} sr^n &= r^{-n}s \\ sr^{25} &= r^{-25}s \\ &= r^{50-25}s \\ &= r^{25}s \end{aligned}$$

Since $sr^{25} = r^{-25}s = r^{25}s$ (because $r^{50} = e$), r^{25} commutes with both r and s and so lies in the centre. By Exercise 2.5 of chapter 7, subgroups of $Z(G)$ are normal subgroups of G, thus K is normal.

Hence, by Proposition 3.7, HB chapter 6 p27, k is a normal subgroup of D_{50}.

e)
All elements in the subgroup H are of the form $r^{2i}s^j$, with the exponent of the r term being even and the element e.

As r^{25} has an odd exponent, its only other element is e.

Thus $H \cap k = \{e\}$.

f)
By Proposition 1.9, HB Chapter 6 p31,

D_{50} has order 100

H has order 50

k has order 2

As $100 = 50 \times 2$, and $H \cap k = \{e\}$, then $D_{50} \cong H \times k$.

Question 6:

Given G is a group of order $48668 = 4 \times 23^3$.

a)
By Definition 1.7, HB Chapter 8 p38, the Sylow 2-subgroups of G are order 4, and the Sylow 23-subgroups of G are order $23^3 = 12167$.

b)
Using The Sylows Theorems, Theorem 1.12 , HB Chapter 8 p39,

For the Sylow 2-subgroups of G;

$$n_2 \equiv 1 \pmod 2$$
$$n_2 \mid 12167 \equiv 1 \pmod 2$$

given the factors of 12167 are $1, 23, 529, 12167$, thus the possible values of n_2 are;

$$n_2 = 1, 23, 529, 12167$$

as all these possible alues are odd, they are all congruent to $1 \pmod 2$.

For the Sylow 23-subgroups of G;

$$n_{23} \mid 4 \equiv 1 \pmod{23}$$

given the factors of 4 are $1, 2, 4$, thus the possible value of n_{23} are;

$$1 \equiv 1 \pmod{23}$$
$$2 \equiv 2 \pmod{23}$$
$$4 \equiv 4 \pmod{23}$$

as 1 is the only value congruent to $1 \pmod{23}$, thus by Lemma 1.9, HB Chapter 8 p39, there is a unique Sylow 23-subgroup of G, which is normal.

Question 7:

a)
Let q be a prime, with $q \neq 29$

A counter example to the proposition that the order of the group $29q$ is cyclic is;

$$q = 2, \text{ with } \mid 29q \mid = 58$$

and thus,

D_{29} is non abelian, thus not cyclic.

b)
Line 7 of the 'proof' is false. Using Definition 3.6, Propositions 3.7 and 3.10, HB Chapter 5 p 27, H and K are cyclic subgroups of G, but being cyclic does not imply being normal. A cyclic subgroup of a non-abelian group need not be normal.

Part B

Question 8: Question 9 from TMA02 material.

Let C[0,1] be the group of all continuous functions from [0,1] to ℝ. Let $+$ be the operation defined By

$f + g$ is the function from $[0, 1]$ to ℝ given by $(f + g)(x) = f(x) + g(x)$

G1 Closure: If $f, g \in C[0, 1]$ then $x \in C[0, 1], f(x), g(x) \in \mathbb{R}$

Since the sum of two continuous functions is continuous, Combination rules for continuous functions from ℝ to ℝ, HB Chapter 13 p65,

$$f + g \in C[0, 1]$$

Hence, $C[0, 1]$ is closed under $+$.

G2 Identity: Let e be the function from $[0, 1]$ to ℝ defined by $e(x) = 0$ for all $x \in [0, 1]$.

Then for any $f \in C[0, 1]$ and $x \in [0, 1]$,

$$\begin{aligned}(f + e)(x) &= f(x) + e(x)\\ &= f(x) + 0\\ &= f(x)\end{aligned}$$

Thus, $f + e = f$ for all $f \in C[0, 1]$, so e is the identity element of $C[0, 1]$.

G3 Inverse: Define $(-f)(x) = -f(x)$, for all $x \in [0, 1]$.

Since f is continuous $-f$ is also continuous, by Combination rules for continuous functions from ℝ to ℝ, HB Chapter 13 p65.

Then

$$(f + (-f))(x) = f(x) - f(x) \qquad = 0(x) = e$$

Thus every element has an inverse in $C[0, 1]$.

G4 Associativity: Associativity holds because we are told in the question that addition in $\mathbb{R}$ is associative.

Hence, $(C[0,1],+)$ is a group.

Question 9: Question 11 from TMA02 material.

a)

$$\mathbb{Z}_3 \times \mathbb{Z}_4 \cong \mathbb{Z}_2 \times \mathbb{Z}_6$$

$$\mathbb{Z}_2 \times \mathbb{Z}_2 \cong \mathbb{Z}_4$$

b)

Group	Order of group	Isomorphic groups	Order of elements	Cyclic? Abelian
$\mathbb{Z}_{15}$	15	$\mathbb{Z}_3 \times \mathbb{Z}_5$	$1, 3, 5, 15$	Cyclic
$\mathbb{Z}_2 \times \mathbb{Z}_{10}$	20	$\mathbb{Z}_2 \times \mathbb{Z}_2 \times \mathbb{Z}_5$	$1, 2, 5, 10$	Abelian, non-cyclic
Dic_6	24	$\mathbb{Z}_2 \times \mathbb{Z}_2 \times \mathbb{Z}_6$	$1, 2, 4, 6, 12, 24$	non-abelian
$\mathbb{Z}_3 \times \mathbb{Z}_4$	12	$\mathbb{Z}_{12}$	$1, 2, 3, 4, 6, 12$	Cyclic, Abelian
$\mathbb{Z}_2 \times \mathbb{Z}_2 \times \mathbb{Z}_3$	12	$\mathbb{Z}_{12}$	$1, 2, 3, 4, 6$	Non-Cyclic